Partial Solutions To Exercises In Topology

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April 11, 2026

Intro

Dr. Lior Kama recommended me this book. It has exercises that I should probably do assuming I want to learn something. I'll probably have to write proofs in English at some point and I figured it might be a great choice to start with it now. Assuming for some reason you're reading this exact same book, you may find those solutions useful (even though I'm not going to solve all of the questions). My packages doesn't really support English really well so i ducttaped something.

Oh and the book appears to be from the 60s so some definitions may be a bit outdated or smth.

Contents

. . . . **(2)** . . . .

Napkin 6.1.3

A metric space M is bounded iff (definition) $\exists D \in \mathbb{R}. \forall p, q \in M \colon d(p,q) \leq D$ iff $\forall p \in M. \exists R_p \in \mathbb{R}. \forall q \in M \colon d(p,q) < R_p$.

Proof. One side is trivial. For the other one, suppose there is some $\mathbb{R} \in R_p$ that bounds $d(p,q)$ for all $q \in M$. Fix some $p \in M$ (assume $M \neq \varnothing$, otherwise every number is trivially is bound), and let $q_1, q_2 \in M$. Then:

$$d(q_1,q_2) \leq d(q_1,p) + d(p,q_2) = d(p,q_1) + d(p,q_2) < 2R_p$$

Hence $2R_p$ is a bound for the distance between each $q_1, q_2 \in M$. ■

Napkin 6.1.6

In metric spaces, totally bound implies boundness.

Proof. Let M be a totally bound space, meaning there are finitely many ε-balls denoted as $U(x_i, \varepsilon_i)$ that covers M. The union of balls is an open set, denotes as U, and each open set U has a basic neighborhood (another ball) containing it. Finally there is an open ball $U(x, \varepsilon)$ that contains the whole of M. In fact, for each $x, y \in M$ from triangle inequality $x, y \in U(x, \varepsilon)$ implies $d(x,y) < 2\varepsilon$. Finally M is bounded. ■

. . . . (3) . . . .

[Fundamental Concepts]

3A ∼ Examples of topologies

1. let $\mathscr{F}$ be the collection of all closed, bounded sets in $\mathbb{R}$. Prove $\mathscr{F}' := \mathscr{F} \cup \{\mathbb{R}\}$ is a family of closed sets.

   Proof. trivially, $\varnothing, \mathbb{R} \in \mathscr{F}'$. Let $A \subseteq \mathscr{F}'$ be a group of closed sets, WOLOG $\mathbb{R} \notin A$. Since A contains closed sets, $\bigcap A \in \mathscr{F}$. Same goes for union. ■

2. let $A \subseteq X$, prove $\mathscr{X} = \{B \in \mathcal{P}(X) \mid A \subseteq B\} \cup \{\varnothing\}$ is a topology.

   Proof. Proof by definition.

   - Immediately $\varnothing, A \subseteq X \in \mathscr{X}$.
   - Let $\mathcal{B} \in \mathcal{P}(X)$. We'll prove that $\bigcup \mathcal{B} \in \mathscr{X}$. Clearly $\bigcup \mathcal{B} \in \mathcal{P}(X)$. We'll prove $A \subseteq B$. Since $\forall B \in \mathcal{B}$ we know $A \subseteq B$ by definition, we get that $A \subseteq \bigcup \mathcal{B}$. Hence $\bigcup \mathcal{B} \in \mathscr{X}$ by definition.
   - Let $B_1 \ldots B_n \in \mathscr{X}$ be a finite number of sets. We'll prove $\bigcap B_i \in \mathscr{X}$. From similar arguments $A \subseteq \bigcap B_i$ and we finished.

   Question: how does interior and closure looks like?

   Assuming $A \subseteq B \in X$, we get $\operatorname{Int} B = B \in \mathscr{X}$ since B is an open set. Given some $B \notin \mathscr{X}$, we know that it contains no open sets (since if $C \subset B$ and $A \not\subset B$ then $A \not\subset C$ therefore $C \notin \mathscr{X}$) apart from $\varnothing$, hence the union of all of the open sets is $\varnothing$.

   For the closure, let some $B \subset X$. We know that $\operatorname{Cl} B = B \setminus B^{\circ}$. Then if $A \subset B$ then $\operatorname{Cl} B = \varnothing$, otherwise $\operatorname{Cl} B = B$.

   For the frontier, we know that $\partial B = \operatorname{Cl} B \setminus \operatorname{Int} B$. Quick check will yield the result that if $A \subset B$ then $\partial B = \varnothing \setminus B = \varnothing$, while if $A \cap B \neq \varnothing$ then $\partial B = B \setminus \varnothing = B$, therefore $\partial B = \operatorname{Cl} B$. Finally:

   $$\operatorname{Int} B = \begin{cases} B & B \in \mathscr{X} \\ \varnothing & B \notin \mathscr{X} \end{cases} \qquad \operatorname{Cl} B = \begin{cases} \varnothing & B \in \mathscr{X} \\ B & B \notin \mathscr{X} \end{cases} \qquad \operatorname{Fr} B = \operatorname{Cl} B$$

   We'll notice that for $A = \varnothing$ we get the discrete topology, and for $A = X$ we get the trivial topology. ■

3B ∼ Frontiers in the plane

I'll proof for $\mathbb{R}$ because I'm lazy but it generalized easily. Let a $A \subseteq \mathbb{R}$ be a closed set. Denote A' some different dense group in $R \setminus A$, such as $(R \setminus A) \cap \mathbb{Q} =: A'$. If $B \subseteq A'$ is closed, and for some $a \in B$ we need to demonstrate $a \in A$. If $a \notin A$ then $a \in R \setminus A$

We notice that $A \subseteq (\mathbb{R} \setminus A')$ and $A \subseteq \operatorname{Cl} A'$

3H ∼ F_σ and G_δ

A subset of a topological space X is a G_δ iff it's a countable itersection of open sets, and an F_σ if it's a countable union of closed sets.

1. The complement of a G_δ is an F_σ and vise versa.

   Proof. We'll proof one side, the other one is trivial from duality. Let $G \subset X$ be some G_δ. We'll proof $\overline{G_\delta}$ is a F_σ. Denote $(G_i)_{i=1}^{\infty}$ be open sets such as $\cup G_i = A$ (exists be definition). From De-Morgan we get:

   $$G^c = \left(\bigcup G_i\right)^c = \bigcap {G_i}^c =: \bigcap F_i$$

   Since G_i is open, then $F_i := \overline{G_i}$ is closed. Hence $\overline{G} = \bigcap F_i$ is a finite intersection of closed sets, then $\overline{G}$ is a F_δ as interned. ■

2. An F_σ can be written as a union of countable $F_1 \subset F_2 \subset F_3 \subset \cdots$ of closed sets.

Proof. Consider some F_σ-set A. Then we have $(\mathcal{F}_n)_{n=1}^{\infty}$ a family of closed sets such that $A = \bigcup_{n=1}^{\infty} \mathcal{F}_n$. Define F_n as the following:

$$F_1 = \mathcal{F}_1 \qquad\qquad F_n = \bigcup_{i=1}^{n} \mathcal{F}_n$$

Clearly $F_n \subset F_{n+1}$. Since a finite union of closed set if closed, F_n is closed. Finally, because $\mathcal{F}_n \subset F_n$ we get that $A = \bigcup^{\infty} \mathcal{F}_n \subset \bigcup^{\infty} F_n$ and because $F_n = \bigcup^{n} \mathcal{F}_i \subset \bigcup^{\infty} \mathcal{F}_i = A$ then $\bigcup^{\infty} F_n \subset A$. Finally $A = \bigcup^{\infty} F_n$ where F_n are closed and $F_n \subset F_{n+1}$ as intended. ■

3. A closed set in a metric space if a G_δ.

   Proof. Suppose A is some closed set in a metric space (X, ρ). For the sake of simplicity I'll assume it's complement is an open disk (and not a union of open disk: but it's enough to look at the base of the topology). So $\overline{A} = U(x, \varepsilon)$. Look at the closed set $U_n = [x, \frac{1}{n}]$: we notice that $\bigcap U_n = U(x, \varepsilon)$. Hence $\overline{A}$ is an F_σ, meaning A is a G_δ.

   The general case can be deduce by taking apart A to be a finite union of simple closed subsets (whom complement is an open disk) which is doable since the closed disks form a base for the family of closed sets in a metric space. ■

4. $\mathbb{Q}$ is F_σ in $\mathbb{R}$.

   Proof. Since $\{r\}$ is closed for each $r \in \mathbb{R}$, and $|\mathbb{Q}| = \aleph_0$, we won (because then we could jusk take the union of the singletons of each element $q \in \mathbb{Q}$).

   Note that $\{r\}$ is closed because for each $a \in \mathbb{R} \setminus \{r\} = \overline{\{r\}}$, we get from density that exists such $b \in \mathbb{R}$ that (WOLOG) $r < b < a$, and then for $\delta = a - b$ we get that $(a - \delta, a + \delta)$ is an open disk (and in fact an open set) in $\mathbb{R} \setminus \{r\}$, so $\mathbb{R} \setminus \{r\}$ is open. ■

. . . . **(4)** . . . .

4A ∼ the Sorgenfrey line

1. Verify that the sets $[x, z)$ for some $z > x$ form a bhood base at x for a topology.

   Proof. Denote by $\mathscr{B}_x := \{[x, z) \mid z \in \mathbb{R}_{>x}\}$. We'll proof via the converse of V-a to V-c.

   V-a By definition, for all x we know $x \in [x, z)$.

   V-b Let $V_1, V_2 \in \mathscr{B}_x$. Hence exists $\mathbb{R} \ni z_1 > z_2$ (WOLOG) such that $V_1 = [x, z_1), V_2 = [x, z_2)$. We get that $V_1 \supset V_2$ hence $V_1 \cap V_2 = V_1$. Finally $\mathscr{B}_x \ni V_3 := V_1 = V_1 \cap V_2$ as intended.

   V-c Let $V \in \mathscr{B}_x$. Choose $V_0 = V$. We need to proof that for each $y \in V_0$ there is some $W \in \mathscr{B}_y$ with $W \subset V$. Denote $V = [x, z_x)$ then we can choose $W = [y, z_x)$ and we notice $[y, z_x) \in \mathscr{B}_y$ by definition, and because $y \in V_0 = V$, we get $x \le y < z_x$ so $W = [y, z_x) \subseteq [x, z_x) = V_0$ as intended.

   By Theorem 4.5 we get that the Sorgenfrey line form a topology over $\mathbb{R}$, denoted as τ, whom its open sets are $A \in \mathcal{P}(X)$ iff A contains some $V \in \mathscr{B}_x$ for all $x \in A$. ■

2. Which intervals on the real line are open sets in the Sorgenfrey topology?

   Proof. We'll define $(\mathbb{R}, \tau)$ a topological space as defined above. Since $\mathscr{B}_x$ is a neighborhood base of the topology that it forms, each open set A is an (in/finite) union of some B from the base ($\mathcal{B} = \bigcup_{x \in X} \mathscr{B}_x$). It clearly follows (technically from compactness) that the open sets are a finite union of bounded $[x, z)$ and unbounded $[x, \infty)$, $[\infty, z)$ and $\mathbb{R}$. ■

3. Describe the closure from each of the following subsets: $\mathbb{Q}, A = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\}, \mathbb{Z}$.

   Proof. We'll prove $\operatorname{Cl} \mathbb{Q} = \mathbb{R}$. Let $x \in \mathbb{R}$, then each basic neighborhood (in $\mathscr{B}_x$) of x meets $\mathbb{Q}$, which is true since $\mathbb{Q}$ is dense in $\mathbb{R}$, hence $x \in \operatorname{Cl} \mathbb{Q}$. From the other side clearly $\operatorname{Cl} \mathbb{Q} \subset \mathbb{R}$. So $\operatorname{Cl} \mathbb{Q} = \mathbb{R}$.

   We'll continue to prove $\operatorname{Cl} A = A \cup \{0\}$. If $x \in A \cup \{0\}$, then each neighborhood $[x, z)$ if $x = 0$ immediately intersects A at infinite amount of points, and if $x \in A$ then $A \ni x \in [x, z)$ so $[x, z)$ meets A. From the other side if $x \in \operatorname{Cl} A$ then each neighborhood intersects A, but if $x \notin A \cup \{0\}$ the set A is discrete so for small enough neighborhoods from archmedianity we get a contradiction.

By similar manners $\mathrm{Cl}\,\mathbb{Z} = \mathbb{Z}$. ■

4F ~ Function Spaces

Consider $\mathbb{R}^I$ where $I = [0,1]$.

1. For each $f \in \mathbb{R}^I$, and F a finite subset of I, we define $U(f,F,\delta) = \{g \in \mathbb{R}^I \mid \forall x \in F\colon\ |g(x) - f(x)| < \delta\}$. Show that $U(f,F,\delta)$ (shorten to $\mathscr{B}_f = \{U(f,F,\delta) \mid \delta > 0, F \subset I \wedge |F| \in \mathbb{N}\}$) form a neighborhood base at f.

   Proof. V-a Since $|f(x) - f(x)| = 0 < \delta$ for each δ, we get that $f \in \mathscr{B}_f$.

   V-b Let $V_1, V_2 \in \mathscr{B}_f$. V_1, V_2 are assigned to some $\delta_1, \delta_2,\ F_1, F_2$, and let $\delta = \min\{\delta_1, \delta_2\}, F = F_1 \cap F_2$. Then $\mathscr{B}_f \ni V_3 = U(f,F,\delta)$ is a neighborhood base that satisfies $V_3 \subset V_1 \cap V_2$.

   V-c Suppose some $V \in \mathscr{B}_x$ with $\delta > 0$. For $V_0 = U(f,F,\frac{\delta}{2})$, from the triangle inequality we get that for each $g \in V_0$ the neighborhood $W = U(g,F,\frac{\delta}{2})$ has the property that for all $h \in W$ and $x \in F$:

   $$\underbrace{|h(x) - g(x)| < \frac{\delta}{2}}_{f \in W = U(g,F,\frac{\delta}{2})} \wedge \underbrace{|g(x) - f(x)| < \frac{\delta}{2}}_{g \in V = U(f,F,\frac{\delta}{2})} \therefore |h(x) - f(x)| < \delta \implies h \in V$$

   So $W \subset V$.

   Hence $\mathscr{B}_f$ is a neighborhood base at f. ■

2. Prove $\mathrm{Cl}\{f\} = \{f\}$.

   Proof. Since each neighborhood of f contains f by definition, $\{f\} \subset \mathrm{Cl}\{f\}$. Ad absurdum there is some $f \neq g \in \mathrm{Cl}\{f\}$, ergo $f(x) \neq g(x)$ for some $x \in I$. Hence $|f(x) - g(x)| > \delta$ for some $\delta > 0$ at that very same x where f, g don't agree. Thus for the set $F = \{x\}$ we get that $g \neq U(f,F,\delta)$ by definition, hence a contradiction. Finally $\mathrm{Cl}\{f\} = \{f\}$ as intended. ■

3. Define $V(f,\varepsilon) := U(f,I,\varepsilon)$. Clearly (it's a relatively similar proof) V form a neighborhood base at f, making $\mathbb{R}^I$ a topological space with topology τ.
4. Comparing the topos from 1 and 3, we get that the topology from 1 is a finer topology and the one on 3, because the the sets $F_x = \{x\}$ for each $x \in [0,1]$, the infinite union:

   $$\bigcup U(f,F_x,\varepsilon) = U(f,I,\varepsilon) = V(f,\varepsilon)$$

   is in the topology from 1 since topos are closed to a generalized union.
5. Suppose now $C(\mathbb{R}^I)$ is the set of continues functions $f\colon I \to \mathbb{R}$. Then the topology from 3 at the subspace $C(\mathbb{R}^I)$ is a topology too. Prove it's metrizable by the metric $\rho(f,g) = \sup|f(x) - g(x)|$.

   Proof. idk this is just double inclusion. ■

4G ~ Nowhere dense sets

TODO A set is *nowhere dense* in X iff its closure contains no nonempty open sets. p is *isolated* within a topology iff $\{p\}$ is open. D is *discrete* in X iff each $d \in D$ has a neighborhood U such as $U \cap D = \{d\}$.

1. In a metric space (X,ρ) without isolated points, the closure of a discrete set in X is nowhere dense.

   Proof. ■

2.

. . . . **(5)** . . . .

5A ~ examples of subbases

1. the family of sets with the form $(-\infty, a)$ together with (b, ∞) form a subbase for the usual topology.

Proof. Denote with $\mathscr{U}$ the family of all finite intersections of $(-\infty, a), (b, \infty)$. We'll proof it forms the usual base for the usual topology. Indeed, for each (a, b) open disk from the base with have $(a, b) = (-\infty, b) \cap (a, \infty)$, and $\varnothing = (-\infty, 0) \cap (0, +\infty)$. On the other hand, for each a, b the intersection of (a, ∞) and (∞, b) is either $\varnothing$ if $a \leq b$ or (b, a) if $a > b$, both are in the usual topology, and the intersection of (a, ∞) and (b, ∞) is just $(\min(a, b), \infty)$ which is too open in the usual topology. Therefore each collection of finite elements from the subbase can be broken into pairs of two sets, whom their intersection is from the usual topology, that is closed for finite intersections, and consequently the intersection of that collection is in the usual topology. ∎

2. Let the family of all straight lines in a plane to be a subbase. Then for each $r \in X$ we can take some two lines such that their intersection is $\{x\}$. Hence the family of all intersections is contained inside the set of singletons of X, which is the base for the discrete topology, the finest topology (so we don't care about any other intersections, and indeed that subbase is a subbase for exactly the discrete topology, and not some larger topology).
3. Now take (a, ∞) to be a subbase for a topology. It's easy to proof that for each finite set of $a_1 \ldots a_n$ and $(a_1, \infty) \ldots (a_n, \infty)$ elements of the subbase, the intersection between all of them is $(\min_n\{a_n\}, \infty)$ which is too in the subbase. Hence the subbase is a base. It's easy to observe that the union of elements from that base=subbase are either from the subbase, or $\mathbb{R}, \varnothing$. So the subbase itself is the topology. Clearly (because the subbase is almost the topology itself) the closed sets are $(\infty, a]$ that are the complement of each element of the subbase=base=topology (apart from $\mathbb{R}$). We get that:

$$\operatorname{Cl} A = (-\infty, \sup A] \qquad\qquad \operatorname{Int} A = (\inf A, +\infty)$$

broadly speaking.

5D ∼ No axioms for subbase

Any family of subsets of a set X is a subbase for some topology on X, and that topology is the smallest topology containing that family. TODO

5F ∼ Second countable and separable spaces

A space X is *second countable* iff X has a countable base. X is separable iff a countable subset D of X exists with $\operatorname{Cl}_X D = X$ (such D said to be *dense* in X).

1. A separable metric space is second countable.

Proof. Let some (X, ρ) be a metric space with topology τ. Suppose A is separable. So there is some countable D with $\operatorname{Cl}_X D = X$. WOLOD $D \neq \varnothing$, since otherwise $\varnothing = \operatorname{Cl}_X D = X$ and each topology over X is finite. Lemma: each open set meets D. Ad absurdum exists some A open and $A \cap D = \varnothing$. Then by definition $\overline{A}$ is closed. Since $A \cap D = \varnothing$ and both $A, D \subset X$, we get $D \subset \overline{A}$. Finally $X = \operatorname{Cl}_X A \subset \overline{A} \subset X$, hence $A = X \supset D$, so A meets D at any of its point – a contradiction.

Now, consider the one-to-one mapping $D \times \mathbb{Q} \ni (d, \varepsilon) \mapsto U(d, \varepsilon)$. Its image will be denoted with $\mathscr{X}$.

Let $x \in X$. Clearly $\mathscr{X} \subset \tau$ (it's a group of open disks). For each $U \in \tau$ open set, it's a union of $\bigcup \mathcal{O}$ where $O \in \mathcal{O}$ is a basic neighborhood (open disk) marked as $U(x_O, \varepsilon_O)$. For each ε_O and $x \in U(x_O, \varepsilon_O)$ and $d \in U(x_O, \varepsilon_O)$ we have a rational q_d^O having the property that $0 < q < \min\{\rho(x, d), \varepsilon_O - \rho(x, d)\}$. For that O we'll define $\mathcal{O}_O := \{U(d, q_d^O) \mid d \in D\}$. We immediately from definition and the triangle inequality have $U(d, q_d^O) \subset U(x_O, \varepsilon_O)$. Moreover, for each $y \in U(x_O, \varepsilon_O)$ and some open P with $y \in U(y, \varepsilon_O^d) \subset U(x_O, \varepsilon_O)$ (same similar contraction), P meets D from the lemma at d. Both of them are in $U(y, \varepsilon_O^y)$ so $\rho(y, d) < \varepsilon_O^y < \varepsilon_O$. So we have some rational $q \in \mathbb{Q}$ that satisfies $0 < \varepsilon_O^y < q < \min\{\rho(x, d), \varepsilon_O - \rho(x, d)\}$, and finally we found a open disk $U(d, q) \in \mathscr{X}$ with $y, d \in U(d, q) \subset U(y, \varepsilon_O^y) \subset U(d, q_d^O)$. Hence $\bigcup \mathcal{O}_O = U(x_O, \varepsilon_O)$. Then we have $\bigcup_{O \in \mathcal{O}} \bigcup \mathcal{O}_O = U$, and $\mathcal{O}_O \subset \mathscr{X}$, so U is a union of items from (the now basis) $\mathscr{X}$, since every open set in τ is given by a union (of union) of elements from $\mathscr{X}$ and $\mathscr{X}$ is contained in τ.

We found a base $\mathscr{X}$ that has one-to-one mapping into $D \times \mathbb{Q}$, which is countable because D is countable and $\mathbb{Q}$ is countable, and $\aleph_0^2 = \aleph_0$. Finally $\mathscr{X}$ is a countable base for (X, ρ), so (X, ρ) is second-countable. ∎

2. Every second-countable space is separable and first countable.

Proof. Let (X, τ) be some second-countable space. Then we have a base countable $\mathscr{X}$.

- ***X* is first countable:** let some $x \in X$, so it has some neighborhood system $\mathscr{U}_x$. As explained in the book, for each $A \in \mathscr{U}_x$ we can take a open space A° having the property $x \in A^\circ \subset A$, and then $\mathscr{U}'_x = \{A^\circ \mid A \in \mathscr{U}_x\}$ is a neighborhood system with open sets. Ergo $\mathscr{U}'_x \subset \mathscr{X} \mapsto \mathbb{N}$, so $\mathscr{U}'_x$ is a countable neighborhood system. Finally we have a countable neighborhood system for each $x \in X$ as intended.

  Note: I essentially needlessly repeated the proof shown from theorem 5.4.

- ***X* is separable:** By AC there is a $F\colon \mathscr{X} \setminus \{\varnothing\} \to X$ choice function. We'll prove $D = \operatorname{Im} F$ is a countable dense set.
  - *Countability:* each $x \in D$ maps to a $U \in \mathscr{X}$ by definition. It's a injective map from D to the countable $\mathscr{X}$, so $|D| \leq |\mathscr{X}| = \aleph_0$ as intended.
  - *Density:* observe some C closed set that covers D. Then $\overline{C}$ is open. Since it's open it has some basic neighborhood $U \in \mathscr{X}$ contained in it. By definition there is $d \in D$ such that $d \in U$, unless $\overline{C} = \varnothing$ (in that case the choice function is undefined). Since C covers D, there are no $d \in U \subset \overline{C}$; hence is must be that $\overline{C} = \varnothing$. From that $C = X$.

    Finally, the only closed set that contains D is X, so $\operatorname{Cl}_X D$ being the intersection of all closed that contains D is X. Because $\operatorname{Cl}_X D = X$, we know X is dense. ■

3. The Sorgenfrey line is first countable and separable.

   Proof.
   - **First countable:** we can easily proof that $\{[x, q) \mid q \in \mathbb{Q}\}$ is a countable base for x.
   - **Separable:** $\mathbb{Q}$ is dense because each $q \in \mathbb{Q}$ meets the open set $[q, p)$ where p is rational bigger than q. We don't even need to proof that: it's being immediately followed by (2) + first countability. ■

. . . . **(6)** . . . .

6A ∼ Example of subspaces

5. The rationals, as a subspace of $\mathbb{R}$, do not inherit discrete topology.

   Proof. Suppose the rationals inherit the discrete topology. Hence for each $q \in \mathbb{Q}$, we have a open set $\{q\}$. By definition of relative topology, $q = \mathbb{Q} \cup \{x - \varepsilon, x + \varepsilon\}$ for some $\varepsilon \in \mathbb{R}$. Hence $\mathbb{Q}$ is discrete, contradiction. ■

6C ∼ Subspaces of separable spaces

3. An open subset of a separable space is separable.

   Proof. Let (X, τ) be a separable topological space, and $A \subset X$ be a space with the relative topology. We have some dense $D \subset X$, such that for each $U \in \tau$ basic neighborhood, $U \cap D \neq \varnothing$. We'll prove $D \cap A$ is dense in A. Indeed, for each U' basic neighborhood in A, we have some basic neighborhood U in X such that $U' = U \cap A$ – which is a neighborhood in X too since τ in closed for a finite intersection. Thus $D \cap (U \cap A) \neq \varnothing$ meaning $D \cap U' \neq \varnothing$, so D meets each basic neighborhood, ergo dense. ■

. . . . **(7)** . . . .

7A ∼ Characterization of Spaces using Functions

Denote with $I_A\colon X \to \{0, 1\}$ the indicator for A in a topological space (X, τ). (when $\{0, 1\}$ inherits has the discrete topology from $\mathbb{R}$)

1. I_A is continues iff A is both open and closed.

   Proof. $\Longrightarrow$ If I_A is continues, then $\{1\}$ is an open set, hence $f^{-1}(\{1\}) = A$ is an open set. $\{1\}$ is a closed set too, hence $f^{-1}(\{1\}) = A$ is a closed set.

$\Longleftarrow$ If A is both open and closed, by brute forcing how f^{-1} applies for each of the options for sets in $\mathcal{P}\{0,1\}$ we can verify that f is continues.

■

2. Prove that X has the discrete topology iff whenever Y is a topological space, and $f\colon X \to Y$ is continues.

Proof. $\Longrightarrow$ If X has the discrete topology, for each (Y, τ) topological space and for each open set $U \in \tau$, we have $f^{-1}(U) \in X$ hence open (each set is open in the discrete topo). By theorem 7.2 f is continues.

$\Longleftarrow$ If for each (Y, τ_Y) topological space, and $f\colon X \to Y$, f is continues, then in particular for τ the discrete topology, $Y = X$ and f the identity, the identity is a homeomorphism and the category topological spaces with continues functions as morphisms, we have $\tau_Y = \tau_X$.

■

7H ∼ Disjoint homeomorphism

Suppose some $(X, \tau_X), (Y, \tau_Y)$ topological spaces such that $X = \biguplus X_n, Y = \biguplus Y_n$ a disjoint union of open sets $X_n \in \tau_X, Y_n \in \tau_Y$, then if $X_n \overset{f_n}{\cong} Y_N$ are homeomorphic, then $X \cong Y$.

Proof. Consider a base for the relative topology at X_n, denoted $\mathscr{X}_n$. Since X_n, X_m are disjoint for each $n \neq m$, the union $\mathscr{X} := \biguplus \mathscr{X}_n$ is a base for X; for each open $U \in \tau$, we know that U can be broken uniquely into $U =: \biguplus U_n$ such that $U_n = X_n \cap U$, and $U_n \in X_n$ is a union of elements from the base $\mathscr{X}_n$. Finally the union of the union of all of these elements form U, hence $\mathscr{X}$ indeed generated (via union) every open set in the topology τ, and each set in $\mathscr{X}$ is open (because the relative topology on X_n is weaker than τ_X). Similarly, we can build a base $\mathscr{Y} = \biguplus \mathscr{Y}_n$ for Y.

Since $f_n\colon X_n \to Y_n$ is homeomorphism, it maps the open sets $\mathscr{X}_n \mapsto \mathscr{Y}_n$. By defining:

$$f\colon X \to Y \qquad f(x) = f_{\iota n \in \mathbb{N}\colon x \in X_n}(x) = \biguplus f_n$$

That is well-defined since X_n, Y_n are a disjoint decomposition of X, Y, we have that f maps $\mathscr{X} \mapsto \mathscr{Y}$, as it's the union of f_n that preserves $\mathscr{X}_n \mapsto \mathscr{Y}_n$. Hence $f\colon X \to Y$ is a homeomorphism, so $X \cong Y$. ■

7L ∼ Linear operators and linear functionals

Let $(X, +_X, \cdot_X, ||\cdot||_X)$, $(Y, +_Y, \cdot_Y, ||\cdot||_Y)$ over rings $\mathbb{C}$, be some normed (and in particular, metric and topological) linear spaces.

A *linear operator* $f\colon X \to Y$ is a map that preserves $\cdot, +$.

An operator Γ is *bounded* iff $\exists M \in \mathbb{C}\colon\ ||\Gamma(x)|| \leq M\,||x||$.

1. A linear operator is bounded iff $\sup ||S_1|| < \infty$ where $||S_1|| := \{||\Gamma(x)|| : x \in X \wedge ||x|| = 1\}$.

Proof. $\Longleftarrow$ Suppose the supremum of $||S_1||$ is ∞ (note that $||\Gamma(x)|| \geq 0$). Ad absurdum Γ is bounded by M. By supremum definition, exists a $x \in X$ where $||x|| > 1$ such that $||\Gamma(x)|| > M$. But $||\Gamma(x)||$ is bounded by $M \cdot ||x|| = M$, hence a contradiction. Thus, Γ must be bounded.

$\Longrightarrow$ Suppose Γ is bounded by some M. So for each $||x|| = 1$ we have $||\Gamma(x)|| \leq M \cdot ||x|| = M$ for some M. Hence, $||S_1||$ is bounded. By the completeness axiom, we know that a supremum for S_1 exists in $\mathbb{C}$, as intended.

■

Note: it can be proven that $||S_1||$ is the supremum (or the maximum the the finite-dimensional case) of the singular values of Γ.

2. TFAE for a linear Γ:
   a) Γ is continues at some $x_0 \in X$
   b) Γ is uniformly continues on X
   c) Γ is bounded

Proof. a→b Suppose Γ is continues at x_0. Note that continuity in the topological sense consolidates with continuity in the metric sense, since basic neighborhood sets in a metric space are open disks (and by using disks instead of neighborhoods, we get the same definition).

Then $\forall \varepsilon > 0. \exists \delta > 0\colon\ ||\Gamma(x_0+\delta) - \Gamma(x_0)|| < \varepsilon$. But we know $\Gamma(x_0+\delta) - \Gamma(x_0) = \Gamma(\delta) = \Gamma(x_1+\delta) - \Gamma(x_1)$ by linearity, hence Γ is continues at all of X, for that very same δ, hence uniformly continues.

b→c Suppose Γ is uniformly continues on X. Then for all $x, y \in X$, for each $\varepsilon \in \mathbb{R}$, we have that $||x-y|| < \delta$ for some δ implies that $|\Gamma(x) - \Gamma(y)| < \varepsilon$. Using linearity, fixing $y = 0_X$, and other nonsense we get:

$$||\Gamma(x)|| = ||\Gamma(x -_X y)|| = ||\Gamma(x) -_Y \Gamma(y)|| < \varepsilon$$

In particular, for $\varepsilon = 1$. Denote $v = \frac{\delta}{2||x||}x$. We know that $||v|| = \frac{\delta}{2||x||}||x|| = \frac{\delta}{2} < \delta$. Therefore $||\Gamma(v)|| < \varepsilon = 1$. Now, we have from linearity:

$$\frac{\delta}{2\,||x||}\,||\Gamma x|| = \left|\left|\Gamma\left(\frac{\delta}{2\,||x||}x\right)\right|\right| = ||\Gamma v|| < 1 \implies ||\Gamma x|| < \frac{2}{\delta} \cdot ||x||$$

So $\frac{2}{\delta}$ is a constant bound for x, as intended.

c→a Assume Γ is bounded. Prove it's continues at some $x_0 \in X$. We'll prove for $x_0 = 0$. We know $\Gamma(0) = 0$ as it's linear. Indeed, by boundness:

$$\exists M \in \mathbb{C}. \forall x \in X\colon\ ||\Gamma(x) - \Gamma(0)|| = ||\Gamma(x)|| \leq M\,||x||$$

Hence, for each $\varepsilon > 0$, we can fix an arbitrary small $||y|| < \frac{\delta}{M}$ (this can be done by exploiting some $||y|| \neq 0$, and inductively subtracting $\frac{y}{2^n}$), that will satisfy the condition for continuity at 0 from the statement above.

■

7N ∼ The group of homeomorphisms

Denote by $H(X)$ the group of homeomorphisms operators for a topological space (X, τ).

1. $H(X)$ is a group, with the composition as a operation, and the identity map as an identity.

   Proof. Indeed, for each $f \in H(X)$ we have $f \circ I = f$. A homeomorphism is a one-to-one mapping with its inverse being a homeomorphism too, hence $f^{-1} \in H(X)$, and $f \circ f^{-1} = I$ by definition. We have associativity for each composition of operators in the general sense, regardless of $H(X)$ being a specific set. Finally $H(X)$ is a group. ■

   Note: the set of continues maps is a monad.

2. Let $X = [0,1] =: I$ and $Y = (0,1)$, and define $\varphi(h) = h|_Y$ for each $h \in H(X)$. Prove $\varphi\colon H(X) \to H(Y)$ is a one-to-one, but there are no homeomorphism form X onto Y.

   Proof. There are no homeomorphisms $f\colon X \to Y$ because if there are such, then $f([0,1]) = (0,1)$, and $[0,1]$ is closed while $(0,1)$ is open, both inherits the same topology from $\mathbb{R}$, hence $(0,1)$ is closed, a contradiction. Meanwhile, clearly h is surjective, but it's injective too; for each $f \in H(Y)$, we know that any extension of f to a $g \in H(X)$ such that $f = g|_Y$, will satisfy $g(0) = \lim_{x\to 0^+} f(x)$ in similarly for $g(1)$, which are defined uniquely by f. ■

. . . . **(8)** . . . .

8A ∼ Projection maps

1. The βth projection π_β is continues and open, although $\pi_1\colon \mathbb{R}^2 \to \mathbb{R}$ is not closed.

   Proof. Let some $X_\alpha\colon I \to X$ and let $\Pi = \prod_{\alpha\in I} X_\alpha$. Fix some $\beta \in I$, we'll prove $\pi_\beta\colon \Pi \to X_\beta$ is open and continues, where X_β has some topology τ_β and Π has the Tychonoff topology τ. Let some $U \in \tau$ open. We know U is a union of the base elements u_x, and that $\pi_\beta(u_x) \stackrel{\text{def}}{=} (u_x)_\beta$ is open. Hence, by continuity:

$$\pi_\beta(U) = \pi_\beta\left(\bigcup u_x\right) = \bigcup \pi_\beta(u_x)$$

LHS is necessarily open since it's a union of open sets. Finally π_β is open.

Now let some $U \in \tau_\beta$ open. We'll prove that $\pi_\beta^{-1}(U)$ is open. Clearly $\pi_\beta^{-1}(U) = \prod U_\alpha$ where $U_\beta = U$ and $U_{\alpha \neq \beta} = X_\alpha$, which is open by definition. So $\pi_\beta^{-1}(U)$ is open for each $U \in \tau_\beta$, ergo π_β is continues.

With that said, $\pi_1 \colon \mathbb{R}^2 \to \mathbb{R}$ is not closed. Denote the group $A = \{(x, y) \in \mathbb{R}^2 \colon xy = 1\}$. Consider $f(x) = \frac{1}{x}$ continues map $f \colon \mathbb{R} \to \mathbb{R}$ and $F((x, y)) = y - f(x)$ continues $F \colon \mathbb{R}^2 \to \mathbb{R}$ (because $(0, f(0))$ is continues, and the constant $(y, 0)$ is continues). Finally $A = F^{-1}(\{0\})$ is closed because $\{0\}$ is closed in the T_1 space $\mathbb{R}$, and F is continues. Even though A is closed, we get that $\pi_\beta(A) = (0, \infty)$ is not closed; a contradiction. ■

2. We'll show that the projection $\pi_2 \colon I \times \mathbb{R} \to \mathbb{R}$ is a closed map.

Proof. Consider some $C \subset I \times \mathbb{R}$ closed group. We'll prove $\pi_2(C)$ is closed. We'll show that $\pi_2(C)$ is sequentially closed which is iff closed in $\mathbb{R}$. Suppose $\pi_2(C) \ni x_n \to x$. Consider $y_n \in I \times \mathbb{R}$ chosen (AC) such that $\pi_2^{-1}(y_n) = x_n$. Since $\pi_2(y_n) = x_n \to x$ and $\pi_1(y_n) \in [0, 1]$, and $[0, 1]$ is compact, we can chose using BW some y_{n_k} subsequence such that $\pi_1(y_{n_k}) \to \ell$. So using some random theorem $y \to (\ell, x)$. Since C is sequentially closed $(\ell, x) \in C$, so $\pi_2((\ell, x)) = x \in C$ too by continuity of π_2, as intended. ■

8B ∼ Separating points from closed sets

1. Let some (X_α, τ_α) be topological spaces. Prove $f_\alpha \colon X \to X_\alpha$ for each $\alpha \in A$, then $\{f_\alpha \mid \alpha \in A\}$ separates points from closed sets and f_α are continues[1] in X iff $\mathscr{B} := \{f_\alpha^{-1}(V) \mid \alpha \in A \wedge V \in \tau_\alpha\}$ is a base for the topology on X.

Proof. $\Longrightarrow$ Suppose the collection $f(A)$ (holy abuse of notation) separates points from closed sets. Since f_α are continues, for all $V \in \tau_\alpha$ open indeed $f_\alpha^{-1}(V)$ is open in the topology of X. Now we'll make sure they actually form a base.

Firstly, we'll prove that $X = \bigcup f_\alpha^{-1}(\tau_\alpha)$ (holy abuse of notation 2). Clearly $\bigcup f_\alpha^{-1}(\tau_\alpha) \subset X$. Let $x \in X$, and fix some closed B. We know that exists some $\alpha \in A$ such that $f_\alpha(x) \notin \overline{f_\alpha(B)}$ for each closed such that $x \notin B$ (exists if the topology is nontrivial). Therefore $f_\alpha(x) \in f_\alpha(B)$.

$\Longleftarrow$ Assume $\mathscr{B}$ is a base for the topology on $\underline{X}$. Let some $x \in X$ and $B \in \mathscr{B}$ such that $x \notin B$. We'll prove that for some $\alpha \in A$ we know $f_\alpha(x) \notin \overline{f_\alpha(B)}$. For the sake of contradiction, assume the opposite. Hence for each closed $B \subset X$, $f_\alpha(x) \notin f_\alpha(B)$.

■

TODO

8D ∼ Closure and Interior in products

Let X and Y be topological spaces with $A \subset X, B \subset Y$.

1. Prove $(A \times B)^\circ = A^\circ \times B^\circ$.

Proof. We know $(x, y) \in (A \times B)^\circ$ iff there is some open O such that $(x, y) \in O \subset A \times B$. Since $A \times B$ is a finite product that inherits the box topology, we notice that there are some O_A, O_B such that $O = O_A \times O_B$. Hence $x \in O_A \subset A$ and $y \in O_B \subset B$. By that $x \in A^\circ \wedge y \in B^\circ$ hence $(x, y) \in A^\circ \times B^\circ$. For the converse, each $(x, y) \in A^\circ \times B^\circ$ we have $x \in O_A \subset A^\circ \wedge y \in O_B \subset B^\circ$ for some open O_A, O_B therefore $O = O_A \times O_B$ is open (2 is a finite number) hence $(x, y) \in O_A \times O_B \subset A \times B$ as intended. ■

Note that this proof does not extends into an infinite product, since for the (sometimes false) inclusion $(\prod A)^\circ \supset \prod A^\circ$ we need (at least) the box topology, and the Tychonoff topology is not coarse enough. For example, in $\mathbb{R}^\mathbb{R}$, the set $A = (-1, 1)$ is open hence $A = A^\circ$, so $\prod A^\circ = \prod A = A^\mathbb{R}$. Meanwhile $(\prod A)^\circ = (A^\mathbb{R})^\circ$ is empty; in the open set $g \in U(f, F, \varepsilon)$, when $F \in \mathcal{P}_{\text{fin}}(\mathbb{R}), \varepsilon > 0$, we have some $\mathbb{R} \ni i \notin F$, for which $g(i)$ could be any number in $\mathbb{R}$. Thus, since $f \in (A^\mathbb{R})^\circ$ iff $f \subset U(g, F, \varepsilon) \subset A^\mathbb{R}$ for some function g, and we know $g(i)$ could be any number in $\mathbb{R}$ including $g(i) \notin A$, we get there aren't any open sets contained in $A^\mathbb{R}$, therefore such f does not exists, and $(A^\mathbb{R})^\circ = \varnothing$, even though $A^\mathbb{R} \neq \varnothing$.

[1] not required in the book. Probably a mistake, since Theorem 8.14 states that f_α must be continues functions

(9)

TODO

(10)

10C ∼ Topology of first-countable spaces

Prove the following for X, Y first-countable spaces:

1. $U \subset X$ is open iff $x_n \to x \in U$ implies that x_n is eventually in U.
2. $F \subset X$ is closed iff whenever $x_n \in F$ and $x_n \to x$ then $x \in F$.
3. $f\colon X \to Y$ is continues iff $x_n \to x \implies f(x_n) \to f(x)$.

Proof. 2. By theorem 10.4 and using the fact that F is closed iff $F = \mathrm{Cl}_X F$.

1. $\implies$ Let U be an open set an let $x_n \to x \in U$. We know that x has some basic neighborhood $x \in O \subset U$. Since $x_n \to x$, at some point n_0 we know that $\forall n \geq n_0\colon x_n \in O \subset U$, hence x_n is eventually in U.

   $\impliedby$ Let $U \subset X$ and suppose $x_n \to x \in U$ implies that x_n eventually in U. AFSOC U is not open. Hence $\overline{U}$ is not closed. By (1) and since X and by first-countability exists some $x_n \in \overline{U}$ that converges to some x even though $x \notin \overline{U}$. Therefore $x \in U$ so x_n is eventually in U, but if for some n_0 we know $x_n \in U$ then $\overline{U} \ni x_n \notin \overline{U}$ and that's a contradiction. Finally U is open.

3. $\implies$ Assume $f\colon X \to Y$ is continues. Then each neighborhood $V \subset Y$ of $f(x)$ has some U neighborhood of x such that $f(U) \subset V$. Let $f(x) \in V$ be a neighborhood of $f(x)$. Exists some $U \subset X$ such that $f(U) \subset V$. Therefore $x \in U$ a neighborhood of x, and we get that x_n eventually meets U, after some n_0. Hence for each $n \geq n_0$ we get that $f(x_n) \in f(U) \subset V$, so x_n is eventually meets V. Ergo $f(x_n)$ eventually meets each neighborhood of $f(x)$ as intended.

   $\impliedby$ Assume $x_n \to x$ implies that $f(x_n) \to f(x)$. We'll prove f is continues. Let $V \subset Y$ be an open set. Now consider $f^{-1}(V) =: U$. Let $x_n \to x$ and assume $x \in U$, it's given that $f(x_n) \to f(x)$ too, and we know that $f(x) \in V$. Because V is open, it's sequentially open, and $f(x_n)$ eventually after some n_0 reaches V. Since $f(x_{n \geq n_0}) \in V$ then $x_{n \geq n_0} \in f^{-1}(V) = U$ so x_n eventually meets U. Thereupon we get that U is open via the converse of (1) and by first countability. Finally $f^{-1}(V)$ is open in X for each V open in Y, hence f is continues as intended. ■

(13)

13B ∼ T_0 and T_1-spaces

1. Any subspace of a T_0 or a T_1 space is respectively a T_0 or a T_1 space.

   Proof.
   - Let X be a T_0 space and consider some $A \subset X$ with the relative topology. Let $x, y \in A$. Exists some $U \subset X$ such that (WOLOG) $x \in U \wedge y \notin U$. Therefore the same applies for $A \cap U$ as intended.
   - If X is a T_1, it's the same proof just with U_1, U_2.

   ■

2. Any nonempty product space if T_0 or T_1 iff each factor space is respectively T_0 or T_1 space.

   proof for T_0 spaces. $\implies$ Assume $\prod X_\alpha$ is a nonempty T_0 product space. Since the product is nontrivial, we can fix some $b_\alpha \in X_\alpha$ for each α in the indexing group A. Now consider the group $B_\alpha = \{(x_\alpha)_{\alpha \in A} \mid \alpha = \beta \vee x_\alpha = \beta_\alpha\}$. We know that $B_\alpha \subset \prod X_\alpha$ Therefore from (1) B_α is T_0. Moreover $\pi_\alpha|_{B_\alpha}$ is a bijection because $X_\alpha \ni x \leftarrow\!\shortmid x'$ where $\pi_\beta(x')$ is b_β if $\beta = \alpha$ and x otherwise. Since we know that projections are open and continues, $\pi_\alpha|_{B_\alpha}\colon B_\alpha \to X_\alpha$ is homeomorphism. Because T_0 is a topological property and B_α is T_0 then $X_\alpha \cong B_\alpha$ is T_1 too as intended.

   $\impliedby$ Assume each factor space $(X_\alpha)_{\alpha \in A}$ is T_0. Let some $x \neq y \in \prod X_\alpha$. We know the existence of some α such that $x_\alpha \neq y_\alpha$. For this α we know that exists some $U_\alpha \in X_\alpha$ such that WOLOG

$x \in U_\alpha \wedge y \notin U_\alpha$. Thereafter we get that $\pi_\alpha^{-1}(U_\alpha)$ is open in X and $x \in \pi_\alpha^{-1}(U_\alpha) \wedge y \notin \pi_\alpha^{-1}(U_\alpha)$ which implies that X is T_0.

■

3. TODO
4. TODO

13C $\sim$ The T_0-identification

Let X be a topological space and $\sim$ defined as $x \sim y \iff \overline{\{x\}} = \overline{\{y\}}$.

- Prove $\sim$ is an equivalence relation.

  Proof.
  - If $x = y$ then indeed $\overline{x} = \overline{y}$ so $x \sim x$.
  - If $x \sim y \wedge y \sim z$ then $\overline{x} = \overline{y} \wedge \overline{y} = \overline{z}$ therefore $x \sim z$.
  - If $x \sim y$ then $\overline{x} = \overline{y}$ thereupon we get $\overline{y} = \overline{x}$ so $y \sim x$.

  Finally $\sim$ is an equivalence relation. ■

- Prove that $X/\sim =: \tilde{X}$ is T_0.

  Proof. Let $[x] \neq [y] \in X/\sim$. Then either $x \notin \mathrm{Cl}\{y\} \vee y \notin \mathrm{Cl}\{x\}$, WOLOG assume the first. Because the closure is closed, $X \setminus \mathrm{Cl}\{y\}$ is an open set. It's obviously saturated since for the natural map $P\colon X \to X/\sim$ and $U =: X/\sim \setminus \{[x]\}$, clearly $P^{-1}(U) = X \setminus \mathrm{Cl}_X\{y\}$, hence U is open. Clearly $[y] \in U \wedge [x] \notin U$, as intended. Finally $X/\sim$ is T_0. ■

13E $\sim$ Accumulation points and condensation points

1. In a T_1-space, a is an accumulation point of A iff each neighborhood of a meets A in an infinite set.

   Proof. $\Longrightarrow$ If a is accumulation point of A, consider some neighborhood U of a that we know that meets A. Let some $y_1 \in U \cap A$, then we can separate x from y_1 with some neighborhood $y_1 \notin U_{y_1} \ni x$. Inductively for each y_n we can look at U_{y_n}, we know that $U_{y_n} \cap A$ is nonempty and has some y_{n+1}, and we can separate y_{n+1} from x using some open neighborhood $y_{n+1} \notin U_{y_{n+1}} \ni x$.

   Ad absurdum $U \cap A$ is finite. Then we can number them $y_1 \dots y_n$. We can find neighborhoods $U_{y_1} \dots U_{y_n}$ as shown above (and if $y_{n+1} \notin A \cap U_{y_n}$ we'll define $U_{y_{n+1}} = U_{y_n}$, which satisfies the condition of being a neighborhood of x without y_{n+1}) although U_{y_n} is a open neighborhood of x that doesn't have any of $y_1 \dots y_n$, even though it meets A at some point y_{n+1} which is not at $U \cap A$, hence a contradiction.

   $\Longleftarrow$ If each neighborhood of a meets A is an infinite set, in particular the intersection is nonempty hence a in an accumulation point.

   ■

2. Trivial
3. Practically we need to prove that $A' \supset (A')'$.

   Proof. From (2) A' is closed. We also know that $\mathrm{Cl}\, A' = A' \cup A''$, but since A' is closed $\mathrm{Cl}\, A' = A'$, hence $A' = A' \cup A''$. Then $A'' \subset A'$ as intended. ■

4. Define inductively $A^{n+1} := (A^n)'$ and $A^1 := A'$. We proved $A^1 \supset A^2 \supset \cdots$. Now we need to prove that for any positive integer n, there is a set $A \subset \mathbb{R}$ such that $A, A^1 \dots A^{n-1}$ are nonempty and $A^n = \varnothing$.

   Proof. For $n = 0$ we define $A = \varnothing$, for $n = 1$ we define $A = \{0\}$ and then $A' = \varnothing$. For any $n > 1$ (define $A^0 = A$): (Partially GPTed) notice the observation $A^{n-1} = \{0\} \uplus \{\frac{1}{n} \mid n \in \mathbb{N}\}$ has the property that $\varnothing = A^n = \{0\}' = (A^{n-1})' = ((A^{n-2})')'$. Now we can "integrate backwards". For each $n - 2 > k \geq 0$ (if exists such k) we'll define:

   $$A^k := \bigcup_{k \in A^{k+1}} \left\{ k + \frac{1}{n} \mid n \in \mathbb{N} \right\}$$

   Notice that since k is finite, we get that $(A^k)' = A^{k+1}$. This defines some $A = A^0$ that satisfies the required property. ■

5. In T_1 spaces, obviously a condensation point is an accumulation point and vice versa (from (1)). Although in non-T_1 spaces it's not guaranteed.

13G ∼ Topological groups

A *topological group* is defined as a T_2 space such as the inversion O and the multiplication m maps are continues. The identity of the group is denoted as e.

1. $m\colon X \times X \to X$ is continues iff for each neighborhood W of xy there are neighborhoods U of x and V of y such that $UV \subset W$.

   Proof. Trivial from the definition of continuity and product topology. ■

2. A group is a topological group iff $c\colon G \times G \to G$ defined as $c(x,y) = x \cdot y^{-1}$ is continues.

   Proof. Clearly O the inversion map is continues since $O = \pi_1 \circ c$ is a composition of continues maps. We also know that m is continues, because the map $f(x,y) = (x, y^{-1})$ is continues because $\pi_1 \circ f = I_G$ which is continues and $\pi_2 \circ f = O$ which is continues too. Finally we get that $m = c \circ f$, and a composition of continues map is continues. ■

3.
   - In $\mathbb{R}$, which regular addition and usual topology, we get a topological group.

     Proof. First of all $O(x) = -x$ is continues because $(a,b) \leftarrow\!\!\!\mapsto (-a,-b)$ both are open (in fact, O is an homeomorphism). Moreover $m(x,y) = x + y$ is continues, because $(x_n, y_n) \to (x,y)$ implies $f(x_n, y_n) = x_n - y_n \to x - y = f(x,y)$ and $\mathbb{R}$ is first countable. Finally addition and inversion are continues as intended. ■

   - Each group with the discrete topology is a topological group.

     Proof. Consider some group G with the discrete topology and some $O\colon G \to G$ inversion and $m\colon G \times G \to G$ multiplication map. O is continues because each operator within the discrete topology is continues. $G \times G$ has the discrete topology too, therefore m is also continues. Finally G is a topological group. ■

   - Let $a, b \in G$ a topological group. Each of the maps:

     $$(1)\ x \to x^{-1} \qquad (2)\ x \to ax \qquad (3)\ x \to xb \qquad (4) x \to axb$$

     It's sufficient to prove that (1) and (2) are homeomorphisms because (4) is composition of (2) and (3) and (3) can be proven from (2) in the group that has $m'(x,y) = m(x,y)$.

     For (1), indeed $O(x) = x^{-1}$ is continues. It's well known that inverse is unique in group theory (because let some $a, b \in G$, we get that if $O(a) = O(b)$ thereupon $O(O(a)) = O(O(b))$ and inversion is assumed to be monoidical so $a = b$, and more generally naturally isomorphic to it). Its inverse is also continues, because $O \circ O = I$ therefore $O = O^{-1}$, hence both O^{-1} is open (it's given that O is open). Finally O is an homeomorphism.

     For (2), fix some $c \in G$ and define $f(x) = c \cdot x$. It's a bijection because if $f(x) = f(y)$ we know $cx = cy$, and by multiplying from the right by c^{-1} we get $c^{-1}cx = c^{-1}cy$ so $x = y$. Also for each $x \in G$ we gave $f^{-1}(x) = c^{-1}x$ because $f(c^{-1}x) = cc^{-1}x = x$. Indeed its injective and surjective.

     Clearly $m|_{\{c\}\times G}$ is continues and a restriction of a continues map is continues, and $\{c\} \times G \overset{r}{\cong} G$, finally $f = r \circ m|_{\{c\}\times G}$. Same goes for $c^{-1} = c^{-1}x$, $\{c^{-1}\} \times G \overset{l}{\cong} G$ and $f^{-1} = l \circ m|_{\{c^{-1}\}\times G}$.

     Finally f is bijection, continues and its inverse is continues too.

4. If $\mathscr{U}$ is a neighborhood base at e, then for all $x \in G$, also $\{xU \mid U \in \mathscr{U}\}$ and $\{Ux \mid U \in \mathscr{U}\}$ are neighborhood bases at x.

   Proof. Let $x \in G$, and let some $\mathscr{U}$ neighborhood base for x. We'll prove $x\mathscr{U}$ is a neighborhood base at x. Consider the map $f(y) = xy$. Clearly $f(e) = x$ and $f(\mathscr{U}) = x\mathscr{U}$. Because we proved f is a homeomorphism, for each neighborhood O at x we know that $f^{-1}(O)$ is neighborhood at e hence we have some $\mathscr{U} \ni U \subset O$. Then $x\mathscr{U} \ni xU = f(U) \subset f(f^{-1}(O)) = O$ so indeed by definition $x\mathscr{U}$ is a basic neighborhood. Similarly for $\mathscr{U}x$. ■

(14)

14A ∼ Examples on regularity and complete regularity

1. Let $A \subset X$ and define $\tau = \{U \in \mathcal{P}(X) \mid A \subset U\} \cup \{\varnothing\}$. We've already proven that the closed sets are the sets that does not contain A is 3A(1). Assuming $A \neq X, \varnothing$, we can find some open $x \in A \subset B \neq X$. Then B^c is closed and $x \notin B^c$. But if X is regular, exists some distinct open U, V such that $B^c \subset U$ and $x \in V$, yet $A \subset U$ by definition (unless $U = \varnothing$, but then $B^c = \varnothing$ hence $B = X$ and we picked $B \neq X$). We know $V \ni x \in A \subset U$ and we get that $\varnothing = V \cap U \ni x$ which is a contradiction. In particular it's not completely regular.

   If $A = \varnothing$ then we get the discrete topology which is regular, since every function $f\colon X \to I$ is continues and in particular $f(A) = 0$ if $x \in A$, otherwise 0 which separates each closed A and $x \notin A$.

   If $A = X$ then we get the trivial topology, then given $x \notin A$ and A closed, it is necessary that $A = \varnothing$ and $x \in \mathbb{R}$ (the trivial case). Now consider the function $f\colon X \to I$ that is constant in 1. Constant maps are continues and indeed $f(\varnothing) \subset \{0\}$.

3. The slotted plane (X, τ) is T_2 but not T_3 (which contradicts some online dude that believes $X \cong \mathbb{R}^2$).

   Proof. First we'll prove Hausdorff. Let some distinct points $y \neq x \in X$. Denote with $\rho\colon X \to \mathbb{R}$ the usual metric in $\mathbb{R}^2$, and with $U(\varepsilon, x)$ a open disk at a point x. Now for $\Delta = \rho(x, y)$ we get that the disks $x \in U(\frac{\Delta}{2}, x)$ and $y \in U(\frac{\Delta}{2}, y)$ are clearly distinct, and a particular (trivial) example of a slotted disk.

   With that said, it's not a T_3 space. Indeed, consider the slotted disk $U = U(\varepsilon, 0) \setminus \{(x, x) \mid x \in \mathbb{R}\}$ for $\varepsilon = 1$. We know that U^c is closed by definition. We know that $S_1 \ni x := (\sqrt{2}, \sqrt{2}) \notin U^c$ (as it's in U), and ad absurdum T_3, then there are open O, V such that $x \in O \wedge U^c \subset V$ can be separated. But O must meet U^c and indeed meet V too, which is a contradiction. ■

4. The looped line is Tychonoff: trivial for each $x \neq y$ that are not in the origin, because $\mathbb{R} \setminus \{0\}$ contains a metrizable topology. If $x = 0$ (WOLOG), then there is some $n > y$ and $0 < \varepsilon < \frac{\rho(x,y)}{2}$ hence the basic neighborhood $(-\varepsilon, \varepsilon) \cup [n, \infty) \cup (\infty, n]$ does not intersect $(y - \varepsilon, y + \varepsilon)$.

14F ∼ Urysohn spaces

A X is a *Urysohn space* iff for each distinct points $x \neq y$ there are neighborhoods $x \in U$ and $y \in V$ such that $\operatorname{Cl} U \cap \operatorname{Cl} V = \varnothing$.

1. Every regular T_1 space is Urysohn and every Urysohn space is Hausdorff.

   Proof. Clearly every Urysohn space is Hausdorff since $U^\circ \subset U \subset \operatorname{Cl} U$, therefore if $\operatorname{Cl} U \cap \operatorname{Cl} V = \varnothing$ then $U^\circ \cap V^\circ = \varnothing$. In particular if U, V are neighborhoods then $x \in U^\circ \wedge y \in V^\circ$ so x, y can be separated with U°, V°.

   Now consider some regular T_1-space. We'll prove its Urysohn. Let $x \neq y \in X$. Since we're in a T_3 space, it's Hausdorff, hence there are open V, U such that $x \in V \wedge y \in U$ yet $U \cap V = \varnothing$. By theorem 14.3(b) we know that there are V', U' basic neighbors of x, y respectively such that $\operatorname{Cl} V' \subset V \wedge \operatorname{Cl} U' \subset U$. Hence $\operatorname{Cl} V' \cap \operatorname{Cl} U' = \varnothing$ while $x \in V' \wedge y \in U'$ as intended. ■

2. Not every Urysohn space is semiregular. For example, the looped line is Urysohn (if $x \neq 0$ it's trivial, otherwise) but not regular: the set $(1, 2] \cup [-2, -1) =: A$ is open, and we have $\operatorname{Int}(\operatorname{Cl} A) = \operatorname{Int}([-1, -2] \uplus [1, 2]) = A$. Even though 2 and A are not separatable with open sets.

   I did not find any example of a non-semiregular $T_{2\frac{1}{2}}$ space.

(15)

15C ∼ Perfectly normal spaces

A space is *perfectly normal* iff exists $f\colon X \to I$ that separates exactly $f^{-1}(0) = A, f^{-1}(1) = B$. A perfectly normal T_1 space is called a T_6 *space*.

1. T_6 iff T_4 and every closed set if a G_δ-set.

Proof. $\implies$ Assume (X, τ) is a perfectly normal space, and assume each closed set if a G_δ set. Consider some A, B disjoint closed set. Because they're G_δ, we have some $(A_n)_{n=1}^\infty,\ (B_n)_{n=1}^\infty$ sequence of open sets, such that $A = \bigcap A_n,\ B = \bigcap B_n$. WOLOG assume $A_n \supset A_{n+1} \land B_n \supset B_{n+1}$ (see 3H(2)).

Since we're in T_4 space, by Urysohn's Lemma, we have the existence of a Urysohn function for each two closed sets. In particular for the closed A and $X - A_n$ we have Urysohn f_n^A such that $f_n^A(A) = 0 \land f_n(X - A_n) = 1$. Similarly we have f_1^B such that $f_n^B(B) = 0 \land f_n^B(X - B_n) = 1$. Now define:

$$f_A(x) = \sum_{n=1}^{\infty} \frac{f_n^A(x)}{2^n} \qquad f_B(x) = \sum_{n=1}^{\infty} \frac{f_n^B(x)}{2^n} \qquad f(x) = \frac{f_A(x)}{f_A(x) + f_B(x)}$$

Before dealing with f, let's prove some properties of $f_A(x)$ (which will obviously will apply to f_B too):

- $\boldsymbol{A = f_A^{-1}(0)}$**:** On one side, consider $x \in A$. Then $f_n^A(x) = 0$ for each $n \in \mathbb{N}$. Therefore $f_A(x) = \sum \frac{0}{2^n} = 0$, and we got $A \subset f_A^{-1}(0)$. On the other side, consider some $x \notin A$. Then $x \notin \bigcap A_n$, hence there's some $k \in \mathbb{N}$ such that $x \notin A_k$. Since $A_k \supset A_n$ for each $k > n$, thereafter $x \in A_{n>k}^c$, we get that:

$$f_A(x) = \sum_{n=1}^{\infty} \frac{f_n^A(x)}{2^n} = \sum_{n=1}^{k-1} \frac{1}{2^n} + \sum_{n=k}^{\infty} \frac{0}{2^n} < 1$$

  So $f_A(x) \neq 1$, and we got $f^{-1}(0) \subset A$. Finally $A = f_A^{-1}(0)$ as intended.
- $\boldsymbol{f_A(x)}$ **is continues:** in the proof for Tietze's extension theorem (15.8), we've shown that a function series of continues maps converges to a continues map.

Similarly $B = f_B^{-1}(0)$

Let $x \in X$. Now we're going to analyze the term $f_A(x) + f_B(x)$:

- If $x \in A$, then $x \notin B$, and we're going to get $f_A(x) + f_B(x) = f_B(x) \leq 1$. Since $f_B(x) = 0$ if and only if $x \in B$, then $f_A(x) + f_B(x) \neq 0$ in that case.
- If $x \in B$, we're going to get similarly that $0 < f_A(x) + f_B(x) = f_A(x) \leq 1$.
- If $x \notin A \land x \notin B$, then neither $f_A(x)$ of $f_B(x)$ are 0, then $f_A(x) + f_B(x) > 0$.

Now we're going to prove that $f\colon X \to I$ separates perfectly A and B.

- $\boldsymbol{f}$ **has the domain** $\boldsymbol{X}$**:** indeed for each $x \in X$ we've shown that $f_A(x) + f_B(x) \neq 0$, so f is well-defined.
- $\boldsymbol{f}$ **has the range** I**:** indeed because $f_A(X), f_B(X) \subset I$, then f gets the maximum value $\frac{1}{1+0} = 1$ and the minimum value $\frac{0}{r} = 0$.
- $\boldsymbol{f}$ **is continues:** an addition and well-defined division of continues functions f_A, f_B is continues too.
- $\boldsymbol{A = f^{-1}(0)}$**:** notice that:

$$f(A) = \frac{f_A(x)}{f_A(x) + f_B(x)} = \frac{0}{0 + \underbrace{f_B(x)}_{\neq 0}} = 0 \qquad f(A^c) = \frac{\overbrace{f_A(x)}^{\neq 0}}{f_A(x) + f_B(x)} > 0$$

- $\boldsymbol{B = f^{-1}(1)}$**:** notice that:

$$f(B) = \frac{\overbrace{f_A(x)}^{\neq 0}}{f_A(x) + 0} = 1 \qquad f(B^c) = \frac{f_A(x)}{f_A(x) + \underbrace{f_B(x)}_{>0}} < 1$$

Finally X is perfectly normal. Moreover, since X is T_4 it's also T_2. Finally X is a T_6 space.

$\impliedby$ Assume X is a T_6-space. We'll prove each closed set in X is a G_δ-set.

Let $A \subset X$ be a closed set. Then $X \setminus A$ is an open set, and since T_6 space is T_3, we have some open U such that $U \subset \operatorname{Cl} U \subset X \setminus A$. Denote $B = \operatorname{Cl} U$ and we get that A and B are disjoint closed sets. By perfect normality, we have a continues $f\colon X \to I$ such that $A = f^{-1}(0)$ and $B = f^{-1}(1)$.

Consider the groups:

$$U_n = \left\{ x \in X \mid f(x) < \frac{1}{n} \right\}$$

Since f is continues and in particular lower semicontinues, then U_n are open. Now clearly $A \subset \bigcap A_n$, but also $\bigcap A_n \subset A$, we know that $\forall n \in \mathbb{N}\colon f(x) < \frac{1}{n}$, hence $f(x) = 0$, which happens iff $x \in A$. Finally the countable open intersection $\bigcap A_n = A$, hence the closed A is G_δ as intended. ■

2. Every metric space is perfectly normal.

Proof. We'll prove each closed set is G_δ. Indeed, for each closed set A, consider the following open sets:

$$U_n = \left\{ x \in X \mid \rho(x, A) < \frac{1}{n} \right\}$$

Those are open sets, since for each $a \in U_n$ the radius $r = \frac{1}{n} - \rho(a, A)$ we clearly get that $r > 0$, therefore it's a neighborhood of A, and that $U(a, r) \subset U_n$.

Now we'll prove $\bigcap U_n = A$. Clearly $A \subset \bigcap U_n$. For each $x \in \bigcap U_n$, ad absurdum $x \notin A$. Then $d(x, A) > 0$ therefore exists some $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < d(x, A)$. This implies that for this n, $x \notin U_n$ which contradicts the assumption $x \in \bigcap U_n$.

Finally, each closed set is G_δ. Hence by (1) a metric space is T_6 and in particular perfectly normal. ■

15D ∼ Retraction and extension

1. A retract in a T_2space is a closed set.

Proof. TODO ■

15G ∼ Extremally disconnected spaces

15H ∼ Hahn-Banach theorem

1. Prove the Hahn-Banach theorem: Let X be a linear space, and a seminorm $p\colon X \to \mathbb{R}$. Let A be a linear subspace of X and f be a linear functional on A such that $\forall x \in A\colon f(x) \leq p(x)$. Then f can be extended to $F\colon X \to \mathbb{R}$ such that $\forall x \in X\colon F(x) \leq p(x)$.

Proof. TODO ■

. . . . (16) . . . .

16F ∼ Cardinality and countability axioms

1. A separable first-countable first-countable space has a cardinal at most $\mathfrak{c}$.

Proof. Consider $A = (a_i)_{i=1}^{\infty}$ be a dense set in X (by separability). Similarly to 10.4, for each $x \in X$, we know that there is $\mathscr{U} = (U_n)_{n=1}^{\infty}$ countable basic neighborhood, and we can define $\mathscr{U}_n = \bigcap_{k=1}^{n} U_n$ to get another countable basic neighborhood. Since A is dense $\mathscr{U}_n$, we know $A \cap \mathscr{U}_n \neq \varnothing$. Then we can build a choice function $x_n\colon \mathbb{N} \to X$ s.t. $x_n \in A \cap \mathscr{U}_n$. By definition $x_n \to x$ and $x_n \subset A$.

Now consider the identification $x \mapsto x_n$. It's injective since for each two different x, y we'll get different $x \leftarrow x_n \neq y_n \to y$ otherwise they would've converge to the same point. Since that mapping is into $x_n \in \mathcal{P}(A)$, we get that $|X| \leq |\mathcal{P}(A)| = 2^{\aleph_0} = \mathfrak{c}$. ■

2. If X is separable then $|C(X)| \leq \mathfrak{c}$.

Proof. Consider some $f \in C(X)$. Then $f\colon X \to \mathbb{R}$ and continues. Consider the injection $C(X) \overset{F}{\mapsto} C(A)$ that maps $F(f) = f|_A$. It's injective, since for each two $f_1, f_2 \in C(X)$, if $f_1|_A = f_2|_A$, then for each $x \in X$ we can take $A \ni x_n \to x$ (as shown in (1)) and by 10.5(c) we get that:

$$f_1(x) \leftarrow f_1(x_n) = f_1|_A(x_n) = f_2|_A(x_n) = f_2(x_n) \to f_2(x)$$

Then clearly $\forall x \in X\colon f_1(x) = f_2(x)$. Finally because $C(A) \subset A \to \mathbb{R}$ we get that $|C(X)| \leq |C(A)| \leq \left|\mathbb{R}^A\right| = \mathfrak{c}^{\aleph_0} = \mathfrak{c}$. ■

3. If (X, τ) is second-countable, then $|\tau| \leq \mathfrak{c}$.

Proof. We know that τ has a countable base $(U_n)_{n=1}^{\infty}$. Therefore each $U \in \tau$ is a union of U_n-s, aka there exists $I \subset \mathbb{N}$ s.t. $U = \bigcup_{i \in I} U_i$. Therefore (AC) the mapping $\tau \mapsto \mathcal{P}(\mathbb{N})$ that maps each open U to such an indexing set I will be clearly injective (union of the same elements yields the same set) and we get that $|\tau| \le |\mathcal{P}(\mathbb{N})| = \mathfrak{c}$. ∎

.... **(17)** ....

17A ∼ Example of compactness

1. An infinite set X with the cofinite topology is compact.

   Proof. We'll prove that the cofinite topology is compact. It would follow that each infinite set is compact, since the infinite sets are closed. Let $\mathscr{U}$ be an open cover of X. Then $\bigcup \mathscr{U} = X$ and $\forall U \in \mathscr{U}: \ |U^c| < \infty$. Fix soem $U \in \mathscr{U}$. Denote $u_1 \dots u_n \in U^c \subset X$ the elements of U^c, then since $X = \bigcup \mathscr{U}$ we must have for each $j \in [n]$ some U_j s.t. $u_j \in U_j$. Then:

   $$X = U \uplus U^c = U \cup \biguplus_{j=1}^{n} \{x_j\} \subset U \cup \bigcup_{j=1}^{n} U_j \subset X$$

   Finally $U, U_1 \dots U_n$ is a finite subset of $\mathscr{U}$ s.t. its union covers X as intended. ∎

2. Which subset of the Sorgenfrey line $\mathbb{E}$ are compact?

   Proof. ok this is hard ∎

17I ∼ σ-Compact spaces

A space X is called *σ-compact* iff it can be written as the union of countably many compact sets. X is hemicompact iff there is a sequence $\mathcal{K} = (K_i)_{i=1}^{\infty}$ of compact subsets of X s.t. for each K compact there is $n \in \mathbb{N}$ s.t. $K \subset K_n$.

1. Every hemicompact space is σ-compact, while the converse fails.

   Proof. Clearly every finite set is compact in each topological space (consider a cover $\mathscr{O}$, then for each x_i in the set we can find $x_i \in O_i \in \mathscr{O}$ and the union of O_i-s). Hence for all $x \in X$ the singleton $\{x\}$ is compact and we can find $K_x \in \mathcal{K}$ s.t. $x \in K_n$. Now:

   $$X = \biguplus_{x \in X} \{x\} \subset \bigcup_{x \in X} K_x \subset \bigcup_{n \in \mathbb{N}} K_n = \bigcup \mathcal{K} \subset X$$

   Therefore $\mathcal{K}$, which is a family of countable open sets, covers X.

   Meanwhile, the converse of the theorem fails. Consider $\mathbb{Q} \subset \mathbb{R}$ with the relative topology inherited from the usual topology. Then it's clearly σ-compact as the singletons $\{q\} \subset \mathbb{Q}$ are compact and $|\mathbb{Q}| = \aleph_0$. It's not hemicompact; a locally dense subset of $\mathbb{Q}$ must not be closed as it has a limit point in $\mathbb{R} \setminus \mathbb{Q}$, meanwhile, AFSOC there is such $\mathcal{K}$. (since $\mathbb{Q}$ is second-countable) we have a (nested) countable neighborhood base $(O_i)_{i=1}^{\infty}$ (e.g. $O_n = \left(-\frac{1}{n} + x, x + \frac{1}{n}\right)$) at some $x \in X$. Now for each $n \in \mathbb{N}$ we know that $O_n \not\subset K_n$ (otherwise K_n will be locally dense) thereafter we can pick $x_n \in O_n \setminus K_n$. Clearly $x_n \to x$. Then $S := \operatorname{Im} x_n \cup \{x\}$ must be compact (a converges sequence must have just one limit point (x), therefore it's closed, and it's bounded by O_1). But if $S_n \subset K_n$ then $x_n \in K_n$ which implies a contradiction. ∎

2. A σ-compact space is Lindelöf.

   Proof. We know that existence of a cover (of X) $\mathcal{K} := (K_n)_{n=1}^{\infty}$ made out of compact sets. Consider an open cover $\mathscr{U}$ of X. In particular, $\mathscr{U}$ covers K_n for each $n \in \mathbb{N}$. Since K_n is compact we have a finite subcover of $U_n \subset \mathscr{U}$, s.t. $\bigcup U_n = \mathscr{U}$. Since $\mathcal{K}$ covers X we know that $X = \bigcup_{n=1}^{\infty} K_n$, and as $K_n \subset U_n$, we conclude that $X = \bigcup \bigcup_{n=1}^{\infty} U_n$. Hence $\mathscr{U}' := \bigcup_{n=1}^{\infty} U_n \subset \mathscr{U}$ is a subcover of $\mathscr{U}$ that covers X. Moreover, it's a countable union of finite sets, hence countable. Finally each cover has a countable subcover and X is Lindelöf. ∎

3. The product of finitely many σ-compact spaces if σ-compact.

Proof. Let $X_1 \dots X_n$ be finitely many σ-compact spaces, that forms a product space X. $\mathcal{K}_n$ will be a compact cover of X_n., and we can enumerate $(K_n^i)_{i=1}^{\infty} = \mathcal{K}_n$ to be its elements. Now:

$$X := \prod_{i=1}^{n} X_i = \prod_{i=1}^{n} \bigcup_{j=1}^{\infty} K_i^j = \bigcup_{x_i \in \mathbb{N}^n} \overbrace{\prod_{i=1}^{\infty} K_i^{x_i}}^{:=C_{x_i}}$$

(Where x_i is a function $x_i \colon [n] \to \mathbb{N}$) From Tychonoff we know that C_{x_i} is compact space (in the relative topology of X), which implies being a compact subspace. Since $|\mathbb{N} \to [n]| = \aleph_0$, we get that X is the union of countably compact subsets, therefore it's a σ-compact. ■

17P ∼ Onto maps of compact spaces

Let X, Y be compact Hausdorff spaces.

1. (With the help of some math.SE user that was willing to help me with the hard part) Exists a compact subset $X_0 \subset X$ s.t. $f(X_0) = Y$ but f maps no closed subsets of X_0 onto Y.

   Proof. Consider the family of closed subsets of X such that their image on f is Y, denoted $\mathscr{X}$. It's partially ordered by $\subset$. Now take a chain $\mathcal{C} \subset \mathscr{X}$. Consider the space $\mathcal{K} = \bigcap_{K \in \mathcal{C}} K$. It's an intersection of closed sets, therefore $\mathcal{K}$ is closed too. Since it's a subsets of a compact Hausdorff space, it's also compact. We'll prove $\mathcal{K}$ is a lower bound for $\mathcal{C}$.

   Clearly, $\forall K \in \mathcal{C} \colon \mathcal{K} \subset K$. Therefore it only remains to prove that $f(\mathcal{K}) = Y$. Indeed, let some $y \in Y$. Then AFSOC $y \notin f(\mathcal{K})$, then $f^{-1}(\{y\}) \cap \mathcal{K} = \varnothing$, hence $\bigcap_{K \in \mathcal{C}} f^{-1}\{y\} \cap K = f^{-1}\{y\} \cap \bigcap_{K \in \mathcal{C}} K = \varnothing$. Since Y is Hausdorff, $\{y\}$ is closed, and because f is continues, $f^{-1}\{y\}$ is closed. This means that for each $K \in \mathcal{K}$, $f^{-1}\{y\} \cap K$ is closed, which gives us a family of closed sets in the compact space X. By FIT there must be a finite subcollection, denoted $\mathcal{C}' \subset \mathcal{C}$, with an empty intersection, otherwise the intersection of the family would be non-empty. Now $\bigcap_{K \in \mathcal{C}'} f^{-1}\{y\} \cap K = \varnothing$ is a finite empty intersection of closed sets. Because $\mathcal{C}$ is a (linearly ordered) chain, then $\mathcal{C}'$ is linearly ordered too, and because it's finite, we can obtain some smallest element $\mathbf{K} \in \mathcal{C}' \subset \mathcal{C} \subset \mathscr{X}$ s.t. $\forall K \in \mathcal{C}' \colon \mathbf{K} = K$. Finally $\bigcap_{K \in \mathcal{C}'} K \subset \mathbf{K}$ which means:

   $$\varnothing = \bigcap_{K \in \mathcal{C}'} f^{-1}\{y\} \cap K = f^{-1}\{y\} \cap \bigcap_{K \in \mathcal{C}'} K = f^{-1}\{y\} \cap \mathbf{K}$$

   Hence $\mathbf{K} \cap f^{-1}\{y\}$ so $Y \ni y \notin f(\mathbf{K})$, but $\mathbf{K} \subset \mathscr{X}$ which implies $f(\mathbf{K}) = Y$, and that's a contradiction.

   Finally, each chain has a lower bound. By Zorn's Lemma there exists some minimal element $X_0 \in \mathscr{X}$. Because X_0 is closed in the Hausdorff space X, it's also compact. Clearly, if there is a proper closed subset $X_1 \subsetneq X_0$ s.t. $f(X_1) = Y$, then $X_1 \in \mathscr{X}$ which will implies that X_0 is not minimal in the relation $\subset$ inside $\mathscr{X}$. That gives as a contradiction. Finally we obtained a compact $X_0 \subset X$ for whom f maps no closed proper subsets onto Y, as intended. ■

17Q ∼ Projective spaces

TODO

. . . . (18) . . . .

18B ∼ Hilbert space

Denote with $\mathbb{H}$ the Hilbert space (sequences $x \colon \mathbb{N} \to \mathbb{R}$ s.t. $\sum x_n^2 < \infty$).

1. The distance function is a metric for $\mathbb{H}$.

   Proof.
   - Assume $d(x, y) = 0$. Since $(x_i - y_i) \geq 0$ and we know that $0 = d(x, y)^2 = \sum (x_i - y_i)^2$ then it must be that $x_i - y_i = 0$, therefore $\forall i \in \mathbb{N} \colon x_i = y_i$ and $x = y$ as intended.
   - Symmetry and positiveness are trivial.

- Let $x, y, z \in \mathbb{H}$. Notice that for all $n \in \mathbb{N}$:

$$(d(x,y))^2 \xleftarrow{n\to\infty} \sum_{i=1}^{n}(x_i - z_i)^2 = \sum_{i=1}^{n}\left((x_i - y_i) + (y_i - z_i)\right)^2$$
$$\overset{(*)}{\leq} \left(\sum_{i=1}^{n}(x_i - y_i)^2 + \sum_{i=1}^{n}(y_i - z_i)^2\right)^2 \xrightarrow{n\to\infty} (d(x,y) + d(y,z))^2$$

Where $(*)$ is true because of Cauchy-Schwartz:

$$\sum_{i=1}^{n}(a_i+b_i)^2 = \sum_{i=1}^{n}\left(a_i^2 + b_i^2\right)+2\sum_{i=1}^{n}a_ib_i \leq \sum_{i=1}^{n}a_i^2+\sum_{i=1}^{n}b_i^2+2\sqrt{\sum_{i=1}^{n}a_i\sum_{i=1}^{n}b_i} = \left(\sqrt{\sum_{i=1}^{n}a_i^2} + \sqrt{\sum_{i=1}^{n}b_i^2}\right)^2$$

■

2. If $x^n \colon \mathbb{N} \to \mathbb{H}$ is a sequence in $\mathbb{H}$, then $x^n \to x$ in $\mathbb{H}$ implies $\forall i \in \mathbb{N}$: $x_i^n \to x_i$ in $\mathbb{R}$, while the converse fails.

   Proof. We'll prove that the functions $\pi_i \colon \mathbb{H} \to \mathbb{R}$ defined as $\pi_i(x) = x_i$ are closed (note that those are not projections from a product topology as shown in (4)). Let $K \subset \mathbb{H}$ to be closed. Then $\mathbb{H} \setminus K$ is open. We'll prove $R \setminus \pi_i(K)$ is open. Take a $r \in \mathbb{R} \setminus \pi_i(K)$. Note that in particular $x = \{r\}^{\aleph_0} \in \mathbb{H}$ has the property $\pi_i(x) = r$, so $x \in \mathbb{H} \setminus K$. Now we can take an ε-disk such that $x \subset U(x,\varepsilon) \subset \mathbb{H} \setminus K$. In that disk, $\sqrt{\sum(x_i - y_i)^2} < \varepsilon$, so $d(\pi_i(x), \pi_i(y)) = x_i - y_i < \varepsilon$ when utilizing the ℓ_1-norm. [wip] ■

18D ∼ Subsets and subgroups of a topological space

18H ∼ Manifolds

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